continues. Instead, let’s guide our gaze into the future of string theory by discussing five central questions string theorists will face as they continue the pursuit of the ultimate theory.

What Is the Fundamental Principle Underlying String Theory?

One overarching lesson we have learned during the past hundred years is that the known laws of physics are associated with principles of symmetry. Special relativity is based on the symmetry embodied in the principle of relativity—the symmetry between all constant-velocity vantage points. The gravitational force, as embodied in the general theory of relativity, is based on the equivalence principle—the extension of the principle of relativity to embrace all possible vantage points regardless of the complexity of their states of motion. And the strong, weak, and electromagnetic forces are based on the more abstract gauge symmetry principles.

Physicists, as we have discussed, tend to elevate symmetry principles to a place of prominence by putting them squarely on the pedestal of explanation. Gravity, in this view, exists in order that all possible observational vantage points are on completely equal footing—i.e., so that the equivalence principle holds. Similarly, the nongravitational forces exist in order that nature respect their associated gauge symmetries. Of course, this approach shifts the question of why a certain force exists to why nature respects its associated symmetry principle. But this certainly feels like progress, especially when the symmetry in question is one that seems eminently natural. For example, why should one observer’s frame of reference be treated differently from another’s? It seems far more natural for the laws of the universe to treat all observational vantage points equally; this is accomplished through the equivalence principle and the introduction of gravity into the structure of the cosmos. Although it requires some mathematical background to appreciate fully, as we indicated in Chapter 5, there is a similar rationale behind the gauge symmetries underlying the three nongravitational forces.

String theory takes us down another notch on the scale of explanatory depth because all of these symmetry principles, as well as another—supersymmetry—emerge from its structure. In fact, had history followed a different course—and had physicists come upon string theory some hundred years earlier—we can imagine that these